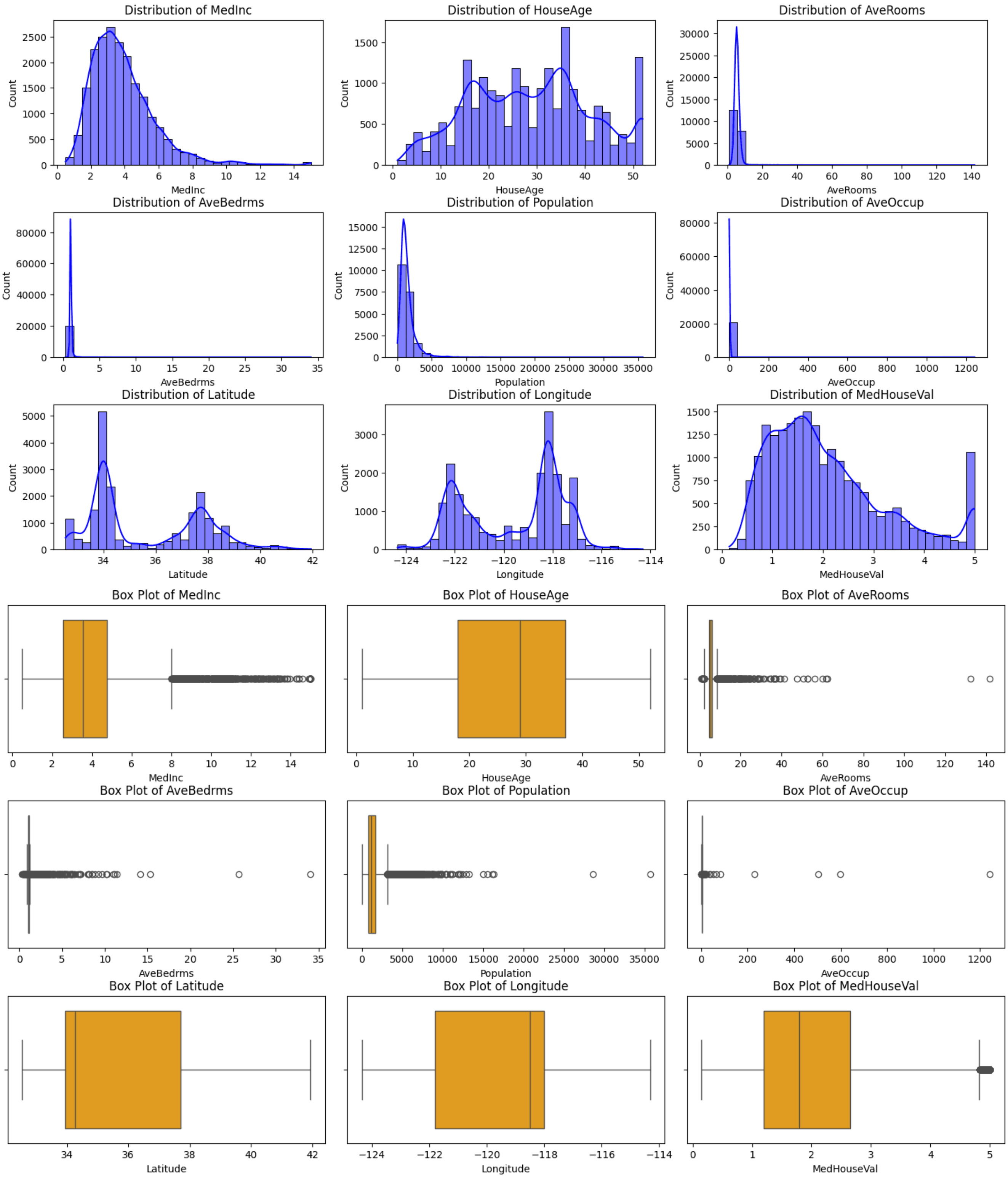
1. Develop a program to create histograms for all numerical features and analyze the distribution of each feature. Generate box plots for all numerical features and identify any outliers. Use California Housing dataset.

PROGRAM:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing

# Step 1: Load the California Housing dataset
data = fetch_california_housing(as_frame=True)
housing_df = data.frame

# Step 2: Create histograms for numerical features
numerical_features = housing_df.select_dtypes(include=[np.number]).columns

# Plot histograms
plt.figure(figsize=(15, 10))
for i, feature in enumerate(numerical_features):
    plt.subplot(3, 3, i + 1)
    sns.histplot(housing_df[feature], kde=True, bins=30, color='blue')
    plt.title(f'Distribution of {feature}')
plt.tight_layout()
plt.show()

# Step 3: Generate box plots for numerical features
plt.figure(figsize=(15, 10))
for i, feature in enumerate(numerical_features):
    plt.subplot(3, 3, i + 1)
    sns.boxplot(x=housing_df[feature], color='orange')
    plt.title(f'Box Plot of {feature}')
plt.tight_layout()
plt.show()

# Step 4: Identify outliers using the IQR method
print("Outliers Detection:")
outliers_summary = {}
for feature in numerical_features:
    Q1 = housing_df[feature].quantile(0.25)
    Q3 = housing_df[feature].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = housing_df[(housing_df[feature] < lower_bound) | (housing_df[feature] >
upper_bound)]
    outliers_summary[feature] = len(outliers)
```

```
    print(f"{feature}: {len(outliers)} outliers")

# Optional: Print a summary of the dataset
print("\nDataset Summary:")
print(housing_df.describe())
```



```
Outliers Detection:
MedInc: 681 outliers
HouseAge: 0 outliers
AveRooms: 511 outliers
AveBedrms: 1424 outliers
Population: 1196 outliers
AveOccup: 711 outliers
Latitude: 0 outliers
Longitude: 0 outliers
MedHouseVal: 1071 outliers

Dataset Summary:

            MedInc      HouseAge  ...    Longitude   MedHouseVal
count  20640.000000  20640.000000  ...  20640.000000  20640.000000
mean       3.870671     28.639486  ...   -119.569704      2.068558
std        1.899822     12.585558  ...      2.003532      1.153956
min        0.499900      1.000000  ...   -124.350000      0.149990
25%        2.563400     18.000000  ...   -121.800000      1.196000
50%        3.534800     29.000000  ...   -118.490000      1.797000
```

Conclusion for individual numerical attribute of the California housing data set

1. MedInc (Median Income)

Distribution is **right-skewed** (positively skewed).

Most households have **low to medium income**, with fewer high-income outliers.

Boxplot shows **several upper-end outliers**.

Income strongly influences house price.

Conclusion:

Higher median income areas generally correspond to higher house values.

2.HouseAge (Median House Age)

 Distribution is slightly **uniform to mildly skewed**.

 Contains capped values (maximum limit in dataset).

 Few outliers compared to income.

Conclusion:

Most houses fall within a mid-range age bracket. Age has moderate impact on pricing.

3.AveRooms (Average Rooms per Household)

- Highly **right-skewed distribution**.
- Significant number of extreme outliers.
- Some districts show unusually high room counts.

Conclusion:

Presence of luxury or unusually large houses creates extreme values.

4. AveBedrms (Average Bedrooms per Household)

Inference:

- Similar to AveRooms but more concentrated.
- Some high-value outliers detected.
- Slight right skewness.

Conclusion:

Bedroom distribution follows room distribution trend but with fewer extreme cases.

5. Population

nference:

- Highly **positively skewed**.
- Few tracts have very large populations.
- Many outliers on upper end.

Conclusion:

Urban clusters create extreme population values.

6.AveOccup (Average Occupancy)

Inference:

- **Extremely right-skewed.**
- **Contains very large outliers.**
- **Some districts have unusually high occupancy ratios.**

Conclusion:

Outliers may represent data irregularities or densely populated housing blocks.

7. Lattitude

Inference:

- **Distribution reflects California's north-south geography.**
- **Slight clustering around major cities.**
- **Few outliers.**

✅ **Conclusion:**

Geographic concentration impacts housing price patterns.

8. Longitude

Inference:

- Reflects California's east-west distribution.

- Some geographic clustering.
- Minimal extreme outliers.

Conclusion:

Location strongly influences house value.

9. MedHouseVal (Target Variable)

Inference:

- Right-skewed distribution.
- Upper cap present (dataset limitation).
- Several high-value outliers.

Conclusion:

Housing prices are not normally distributed; they are concentrated in mid-range with expensive outliers.

Most outliers found in: AveRooms, AveBedrms, Population, AveOccup

Fewer outliers in: HouseAge, Latitude, Longitude

Target variable also has capped extreme values